CCNA Project Presentation Script

Multi-Branch Enterprise Network Implementation Using Cisco Packet Tracer

Slide 1 – Title

What to say:

“Good morning sir/madam.

My project title is **Multi-Branch Enterprise Network Implementation Using Cisco Packet Tracer**.

The main purpose of this project is to design a company network where the head office, branch office and remote office can communicate with each other.

In this project, I have implemented different networking concepts like DHCP, DNS, OSPF, RIP, BGP, NAT and SSH.”

Slide 2 – How This Project Supports Industries

What to say:

“This project represents a real-world company network.

A company can have a head office, different branch offices and remote offices. All these offices need to communicate with each other.

DHCP automatically gives IP addresses to devices, DNS helps users access resources using names, and routing protocols help routers communicate with each other.

NAT is used for communication with an external network, and SSH provides secure remote access to network devices.

This type of network can be used in IT companies, banks, educational organizations, healthcare and multinational companies.”

Slide 3 – Let's Begin With Configuration

What to say:

“Now I will explain the configuration part of my project.

First, I configured the routers and their interfaces. After that, I configured DHCP and DNS.

Then I configured routing protocols like OSPF, RIP and BGP.

Finally, I configured NAT and SSH and tested the network.”

Slide 4 – Configure R1 HQ

What to say:

“First, I configured **R1 as the Head Office router**.

This router is responsible for connecting the Head Office network with the rest of the company network.

I configured its IP addresses and interfaces and prepared it for routing and DHCP.”

Slide 5 – Configure R2 CORE

What to say:

“Next, I configured **R2 as the Core router**.

The Core router is important because it connects different parts of the company network.

It also participates in routing and connects towards the ISP.”

Slide 6 – R2 Core DCE Interface

What to say:

“Here I configured the serial DCE interface of the Core router.

Because this is a serial connection, the DCE side needs a clock rate.

So I used a clock rate of **64000**.

This helps the serial connection between the routers to work properly.”

Slide 7 – R3 Branch DCE Interface

What to say:

“Here I configured the serial interface of the Branch router.

Since this side is the DCE side, I configured the clock rate as 64000.

This provides timing for the serial communication between the routers.”

Slide 8 – R4 Remote DCE Interface

What to say:

“Here I configured the serial connection for the Remote Office router.

Again, because this is the DCE side, I configured the clock rate as 64000.

After this configuration, I can check whether the serial connection is up.”

Slide 9 – Configure ISP

What to say:

“After configuring the internal routers, I configured the ISP router.

The ISP represents the external network or Internet.

The Core router communicates with this ISP router using BGP.”

Slide 10 – DHCP on R1 – HQ

What to say:

“Now I configured DHCP on the Head Office router.

DHCP automatically provides IP addresses to computers.

So instead of manually entering an IP address on every computer, the computer can request an IP address from the DHCP server or router.

This saves time and reduces manual configuration errors.”

Slide 11 – DHCP on R3 – Branch

What to say:

“Here I configured DHCP for the Branch Office.

When a new computer is connected to the Branch network, it can automatically receive an IP address, subnet mask and gateway.

This makes the network easier to manage.”

Slide 12 – DHCP on R4 – Remote

What to say:

“Similarly, I configured DHCP for the Remote Office.

The devices in the Remote Office can automatically receive their network configuration.

So DHCP is implemented in the different office networks.”

Slide 13 – DNS Server IP Configuration

What to say:

“Next is DNS.

First, I configured the IP address of the DNS server.

DNS stands for **Domain Name System**.

Its main purpose is to convert a domain or host name into an IP address.”

Slide 14 – DNS Server Services

What to say:

“After assigning the IP address, I enabled the DNS service.

For example, instead of remembering an IP address, a user can access a resource using its name.

So DNS makes accessing network resources easier for users.”

Slide 15 – R1 HQ OSPF

What to say:

“Now I configured **OSPF** on the Head Office router.

OSPF stands for **Open Shortest Path First**.

In this project, I used **OSPF Area 0** for the internal network.

OSPF allows routers to dynamically exchange routing information instead of manually configuring every route.”

Slide 16 – R2 Core and /30 Subnets

What to say:

“Here I configured the Core router and used /30 subnets for the point-to-point serial connections.

A /30 network contains four addresses.

The first address is the network address, the next two are usable host addresses, and the last address is the broadcast address.

For example, in 10.0.0.8/30, the usable addresses are .9 and .10.

So /30 is useful for router-to-router connections because we only need two usable IP addresses.”

Slide 17 – R3 Branch OSPF

What to say:

“Here I configured OSPF on the Branch router.

The Branch router participates in OSPF Area 0 so that it can exchange routing information with the other OSPF routers.

This helps maintain connectivity inside the network.”

Slide 18 – R4 Remote OSPF

What to say:

“Here I configured OSPF on the Remote router.

The Remote Office can therefore participate in the internal routing process.

If the network changes, OSPF can update the routing information dynamically.”

Slide 19 – R1 HQ RIP

What to say:

“After OSPF, I configured another routing protocol called **RIP**.

RIP stands for **Routing Information Protocol**.

It is a distance-vector routing protocol.

Here RIP is used as part of the routing configuration in my project."

Slide 20 – R2 Core RIP

What to say:

"Here I configured RIP on the Core router.

The Core router can exchange routing information with the routers participating in RIP.

This demonstrates how different routing protocols can be configured in an enterprise network."

Slide 21 – R3 Branch RIP

What to say:

"Here I configured RIP on the Branch router.

The purpose is to allow the Branch network to learn routes through RIP.

I can verify the learned routes using the routing table."

Slide 22 – R4 Remote RIP

What to say:

"Here I configured RIP on the Remote router.

Now the Remote network can exchange routing information through RIP.

So in this project, I demonstrated both OSPF and RIP routing."

Slide 23 – BGP Between Core and ISP

What to say:

"Now I configured **BGP** between the Core router and the ISP.

BGP stands for **Border Gateway Protocol**.

Unlike OSPF and RIP, BGP is mainly used for routing between different autonomous systems.

In my project, the company network communicates with the ISP using BGP."

Slide 24 – BGP Verification

What to say:

“Here I verify the BGP configuration.

I can use the BGP summary command to check the neighbor relationship between the Core router and the ISP.

If the BGP neighbor is established, it means the BGP connection is working.”

Slide 25 – Configure NAT

What to say:

“Next, I configured **NAT**, which stands for Network Address Translation.

NAT allows private internal IP addresses to communicate with an external network.

In this project, the internal interfaces are translated before traffic goes towards the ISP.”

Slide 26 – NAT Inside and Outside

What to say:

“Here I identified the interfaces as **NAT inside** and **NAT outside**.

The interfaces facing the internal company network are marked as inside.

The interface connected towards the ISP is marked as outside.

This tells the router where the translation should take place.”

Slide 27 – Default Route

What to say:

“Here I configured the default route.

The purpose of the default route is to tell the Core router where to send traffic when it doesn't have a specific route for the destination.

In this project, the ISP is used as the default path for external traffic.”

Slide 28 – SSH on R1

What to say:

“Now I configured **SSH** on R1.

SSH stands for Secure Shell.

It allows me to remotely access and manage the router securely.

Unlike basic Telnet, SSH encrypts the communication.”

Slide 29 – SSH on R2

What to say:

“Here I configured SSH on the Core router.

I created a username and password, generated the required RSA key and enabled SSH on the VTY lines.

Now an administrator can securely access the router remotely.”

Slide 30 – SSH on R3

What to say:

“Here I configured SSH on the Branch router.

This provides secure remote management without physically going to the router.”

Slide 31 – SSH on R4

What to say:

“Finally, I configured SSH on the Remote router.

So secure remote management is available on the routers used in my project.”

Slide 32 – Challenges Faced

What to say:

Solutions and Troubleshooting

- **IP Addressing:** Created an IP addressing plan for each office and assigned unique subnets to avoid conflicts.
- **Serial Connectivity:** Configured serial interfaces using the correct IP addresses, enabled interfaces using no shutdown, and checked DCE clock settings where required.
- **Routing Protocols:** Configured routing protocols and verified route exchange using commands such as show ip route and show ip protocols.
- **DHCP and DNS:** Configured DHCP pools with the correct gateway and DNS details, and tested domain name resolution using ping.
- **NAT:** Configured access control lists, NAT inside and outside interfaces, and verified translations.
- **Network Troubleshooting:** Used commands such as ping, traceroute, show ip interface brief, and show arp to identify and resolve connectivity issues.
- **SSH:** Configured domain names, local user accounts, RSA keys, and VTY lines to enable secure remote access.

Final Conclusion

What to say:

“To conclude, this project helped me understand how a real enterprise network can be designed and configured using Cisco Packet Tracer.

I learned how different offices can communicate using routing protocols, how DHCP can automatically assign IP addresses, how DNS provides name resolution, how NAT provides external connectivity, and how SSH provides secure remote management.

I also learned that proper IP addressing and troubleshooting are very important for maintaining network connectivity.

Thank you.”

If Teacher Asks Questions

Q1. Why did you use DHCP?

Answer:

“DHCP automatically assigns IP addresses to devices. It reduces manual configuration and makes it easier to add new devices to the network.”

Q2. What is DNS?

Answer:

“DNS converts a domain or host name into an IP address. It allows users to use names instead of remembering IP addresses.”

Q3. What is OSPF?

Answer:

“OSPF is a dynamic routing protocol. It allows routers to exchange routing information and find suitable paths through the network.”

Q4. What is RIP?

Answer:

“RIP is a distance-vector routing protocol. It uses hop count as its routing metric.”

Q5. What is BGP?

Answer:

“BGP is a routing protocol used mainly between different autonomous systems. In my project, I used it between the Core router and ISP.”

Q6. What is NAT?

Answer:

“NAT translates private IP addresses into a public-facing address so internal devices can communicate with an external network.”

Q7. Why did you use /30 for serial links?

Answer:

“A point-to-point router connection only needs two usable IP addresses, so /30 is suitable because it provides two usable addresses.”

Q8. What is DCE?

Answer:

“DCE is the side of a serial connection that provides clocking. That's why I configure a clock rate on the DCE interface.”

Q9. Why did you use SSH?

Answer:

“SSH provides secure encrypted remote access to the router. It is safer than using an unencrypted remote-access method.”

Q10. What was your biggest challenge?

Answer:

“My main challenge was troubleshooting connectivity and making sure IP addressing, serial connections and routing protocols were configured correctly. I solved the issues step by step by checking interfaces, routing tables and connectivity.”

Q11. How did you test your project?

Answer:

“I used commands such as ping, show ip route, show ip ospf neighbor, show ip bgp summary, show ip nat translations, and DHCP/DNS tests to verify different parts of the network.”

Q12. What did you learn from this project?

Answer:

“I learned how to build a multi-office network, configure routers and serial links, implement dynamic routing, configure DHCP and DNS, use NAT for external connectivity, and provide secure remote access using SSH.”